Chapter 1: Blood Fundamentals

1.1 The Cellular Composition of Human Blood

Human blood is a specialized body fluid that performs critical transport and protective functions.

The total blood volume in an adult human averages between 4.5 and 5.5 liters, accounting for ~7% body weight.

Plasma makes up roughly 55% of total blood volume, consisting of 90% water, electrolytes, and vital proteins.

Erythrocytes (red blood cells) constitute approximately 45% of blood volume, known as the hematocrit.

1.2 Oxygen Transport and Hemoglobin Kinetics

Each erythrocyte contains approximately 270 million hemoglobin molecules.

Hemoglobin is a tetrameric protein composed of two alpha and two beta globin chains with heme groups.

Oxygen binds reversibly to the ferrous iron (Fe2+) at the center of each protoporphyrin ring.

The sigmoidal oxygen-hemoglobin dissociation curve demonstrates cooperative binding kinetics.

Shift to the right (Bohr effect) occurs with increased hydrogen ion concentration, CO2, and temperature.

Chapter 2: Transfusion Safety

2.1 Compatibility and Crossmatching

Pre-transfusion testing ensures complete immunological compatibility between donor and recipient.

ABO blood group system antibodies (anti-A and anti-B) are naturally occurring IgM antibodies.

An acute hemolytic transfusion reaction can occur with as little as 10 to 15 mL of ABO-incompatible red cells.

Immediate signs include fever, chills, hemoglobinuria, dyspnea, hypotension, and acute renal failure.

2.2 Storage Lesion and Quality Metrics

Whole blood and packed red blood cells are stored at 1 to 6 degrees Celsius in regulated preservation solutions.

CPDA-1 anticoagulated blood has a licensed shelf-life of 35 days, whereas additive solutions extend to 42 days.

Storage lesion changes include progressive decline in 2,3-DPG, intracellular potassium leakage, and microparticles.

Stringent quality controls mandate 24-hour post-transfusion in vivo erythrocyte survival greater than 75%.

Chapter 3: Summary and Review

3.1 Key Clinical Pearls

1. Always confirm two independent patient identifiers before administering any blood component.
2. O Negative uncrossmatched red blood cells should be reserved for catastrophic exsanguinating hemorrhage.
3. RhD negative women of childbearing age must strictly receive RhD negative components to prevent isoimmunization.
4. Post-transfusion hemoglobin typically increments by approximately 1 g/dL per unit of packed red cells.

3.2 Completion Notice

Congratulations on completing this study guide.

You have fulfilled the required reading material for this topic.

Proceed to the Knowledge Quiz to test your retention and earn course points and badges.